FINITE LOCALLY FREE SHEAVES WITH ISOMORPHIC ENDOMORPHISM ALGEBRAS DIFFER BY AN INVERTIBLE TWIST

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument. Two steps are described by the author as a local calculation after reducing to the affine finite free case; that reduction is stated explicitly in the counted span and is the author's own routine verification, not an omission of load-bearing mathematics.

Editorial note. The label lemma-end-unique-up-to-invertible occurs in no other staged Stacks chapter, so no cross-chapter restatement had to be weighed for this card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Finite locally free sheaf of O_S-modules of positive rank. Same chapter, the notion of a finite locally free sheaf of O_S-modules and its rank. Supplies the sheaves F and G, and allows the local reduction to finite free modules over an affine base used in the proof.
- Sheaf of homomorphisms and the endomorphism algebra of a finite locally free sheaf. Same chapter, SheafHom_O(F, F) with its structure of O_S-algebra. Supplies the algebras whose isomorphism is the hypothesis, and the ambient sheaf SheafHom(F, G) inside which the invertible sheaf L is constructed.
- Conjugation by a local isomorphism and locally free sheaves of rank one. Same chapter, the action of a local isomorphism varphi : F -> G on endomorphism algebras by conjugation, and invertibility of a locally free sheaf of rank one. Supplies the condition selecting the generators of L, and the target property that L is invertible.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{F}$ and $\mathcal{G}$ be finite locally free sheaves of $\mathcal{O}_S$-modules of positive rank. If there exists an isomorphism $\mathcal{H}om_{\mathcal{O}_S}(\mathcal{F}, \mathcal{F}) \cong \mathcal{H}om_{\mathcal{O}_S}(\mathcal{G}, \mathcal{G})$ of $\mathcal{O}_S$-algebras, then there exists an invertible sheaf $\mathcal{L}$ on S such that $\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{L} \cong \mathcal{G}$ and such that this isomorphism induces the given isomorphism of endomorphism algebras.*

Proof. Fix an isomorphism $\mathcal{H}om_{\mathcal{O}_S}(\mathcal{F}, \mathcal{F}) \to \mathcal{H}om_{\mathcal{O}_S}(\mathcal{G}, \mathcal{G})$. Consider the sheaf $\mathcal{L} \subset \mathcal{H}om(\mathcal{F}, \mathcal{G})$ generated as an $\mathcal{O}_S$-module by the local isomorphisms $\varphi : \mathcal{F} \to \mathcal{G}$ such that conjugation by φ is the given isomorphism of endomorphism algebras. A

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local calculation (reducing to the case that $\mathcal{F}$ and $\mathcal{G}$ are finite free and S is affine) shows that $\mathcal{L}$ is invertible. Another local calculation shows that the evaluation map

$$\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{L} \longrightarrow \mathcal{G}$$

is an isomorphism. $\square$